The Ouroboros Thesis

Axiom-Free Conditional Uniqueness of the Lutar Invariant: A Machine-Verified Trust Foundation for Governed Agentic AI

Stephen P. Lutar Jr.[1]

[1]SZL Holdings ORCID: 0009-0001-0110-4173

Thesis v24 "Axiom-Free Conditional Uniqueness" — June 8, 2026

Concept DOI (always-latest): 10.5281/zenodo.19944926

Abstract

Governed deployment of agentic artificial intelligence in regulated and defense settings requires more than benchmark accuracy: it requires *checkable* guarantees about what a system did, why it was permitted to do it, and whether the record can be tampered with after the fact. This is **v24** of the Ouroboros thesis program of SZL Holdings, a DOI-pinned lineage (v1–v23) whose subject is a machine-verified trust substrate resting on three pillars: (i) the *Ouroboros loop*, a bounded, well-founded self-governing computation in which "the loop is the product"; (ii) the *Lutar invariant* $\Lambda_k(x) = \left(\prod_{i=1}^{k} x_i\right)^{1/k}$, an equal-weight geometric-mean trust aggregator; and (iii) a *proof-trail / receipt architecture* formalized in Lean 4. The headline advance of v24 is an **axiom-free conditional Λ-uniqueness theorem**. In v23 the conditional uniqueness of Λ was gated on a *declared* project axiom (A6′ block-consistency). Here we report `Lutar.Round13.lambda_unique_of_separable`: under {A1, A2, A3, A5} together with *slice-multiplicativity* (separability), Λ is the unique aggregator, with `#print axioms` = `{propext, Classical.choice, Quot.sound}`— i.e. **no project axiom**, kernel-clean, CI-green, merged to `main` at `b910c276`. This strictly strengthens v23: the extra hypothesis is now a *checkable property of* Φ (weaker than the old *Factors* premise) discharged through Mathlib and already-proved in-tree lemmas, rather than a declared idealization. Crucially, the *unconditional* uniqueness of Λ under {A1–A5} remains machine-checked **false** (in-tree `maxAgg_ne_Lambda`; `maxAgg` and `min` are A1–A5 counterexamples), so Λ stays **Conjecture 1** unconditionally — never a theorem. We also report a separate *experimental, CI-green* tier of frontier theorems (CF-13, CF-17, Wave-13 replay/quorum/HM results, and the Wave-14 pack CF-18/19/20/21), all `#print axioms` -clean, that is never folded into the locked-five. The honesty doctrine is binding throughout: locked-proven = exactly five {F1, F11, F12, F18, F19}; declared axioms disclosed; supply-chain posture SLSA L1 honest with L2 a roadmap (correcting a v23 overstatement); nothing fabricated. We argue that replacing a declared axiom with a checkable property is an *epistemic upgrade*, and that this calibrated honesty is what makes every positive claim credible to an auditor.

Keywords: governed agentic AI, trust aggregation, Lutar invariant, geometric mean, slice-multiplicativity, functional-equation uniqueness, Lean 4, Mathlib, axiom-free verification, `#print axioms`, receipt chain, SLSA, Byzantine consensus, philosophy of mathematics.

Honesty note (verbatim). The Lutar invariant Λ is **Conjecture 1** unconditionally and is *never* claimed proven unconditionally; unconditional uniqueness under A1–A5 is machine-checked *false*. The v24 conditional uniqueness theorem `lambda_unique_of_separable` holds under {A1,A2,A3,A5} + slice-multiplicativity with *no project axiom* (`#print axioms = {propext, Classical.choice, Quot.sound}`). Locked/proven = exactly five formulas {F1,F11,F12,F18,F19} at `c7c0ba17`. The experimental CI-green tier (1323 decl / 23 axioms / 307 sorries at `b910c276`) is a *separate* tier and is never folded into the locked-five. SLSA **L1 honest, L2 roadmap** — *not* L2-verified, L3, FedRAMP, Iron Bank, or CMMC. No fabricated results; no fake citations; 0 runtime CDN.

Contents

1 Introduction 4

2 The honesty doctrine 5

3 The Ouroboros loop: bounded self-governance 6

4 The Lutar invariant Λ and its uniqueness boundary 6
- 4.1 Definition and the aggregation axioms . . . 6
- 4.2 The unconditional refutation: FALSE, machine-checked . . . 7
- 4.3 The old route: uniqueness given factorization, and the A6′ gate . . . 7
- 4.4 The v24 advance: axiom-free conditional uniqueness (CUT-2) . . . 8
- 4.5 Λ is Conjecture 1 — the precise claim structure . . . 9
- 4.6 Worked examples . . . 9

5 The proof-trail / receipt architecture 9
- 5.1 Structural correctness (sorry-free) . . . 10
- 5.2 Security under declared idealizations (axiom-gated) . . . 10

6 The locked kernel and the frontier theorem families 10
- 6.1 The locked five (kernel-verified at `c7c0ba17`) . . . 10
- 6.2 The experimental CI-green tier (at `b910c276`) . . . 10
- 6.3 CF-13 — DEQ input-Lipschitz well-posedness margin . . . 11
- 6.4 CF-17 — floating-point summation error bound . . . 11
- 6.5 Wave-13 — replay completeness, quorum shadow, HM bottleneck . . . 11
- 6.6 Wave-14 — CF-18/19/20/21 (nine kernel-clean theorems) . . . 11

7 Verification status, in detail 12
- 7.1 The wave ledger (Waves 11–14) . . . 12
- 7.2 The drift gate and the canonical numbers . . . 12
- 7.3 The `#print axioms` discipline . . . 12
- 7.4 SLSA supply-chain posture — correcting a v23 overstatement . . . 13

8 Why this is groundbreaking — the honest case 13
- 8.1 Honest comparison to prior art . . . 14

9 Philosophical foundations: slice-multiplicativity as an epistemic upgrade 14
- 9.1 From a declared axiom to a checkable property . . . 14
- 9.2 When is a conditional legitimate? . . . 14
- 9.3 Slice-multiplicativity vs. block-consistency . . . 15

9.4 Steelman objections (with honest rebuttals) . . . 15

10 Empirical posture (measured, not proven) 15

11 Limitations and honesty 16

12 Conclusion 16

A Reproducibility and artifact manifest 17
A.1 Pinned commits and artifacts . . . 17
A.2 How to verify . . . 17
A.3 Trusted computing base . . . 17

B Notation and glossary 18

1 Introduction

Agentic AI systems — models that plan, call tools, retrieve, and act over many steps — are entering domains where the cost of an undetected error is not a bad recommendation but a regulatory breach, a safety incident, or a compromised mission. In such regulated and defense settings the operative question is no longer "how accurate is the model on a benchmark?" but "can the operator *prove*, after the fact and to a skeptical third party, that an action was authorized, that the authorization rested on checkable evidence, and that the record has not been altered?" This is the problem of *governed AI*: an execution substrate in which trust is *earned by checkable evidence* rather than asserted by a vendor. The verifiable-claims program argues that the industry default — assert safety, keep the evaluation private, log to a mutable store — is structurally inadequate [6], and policy bodies have echoed the demand that developers be able to *prove* properties of their systems [18].

This paper is **v24** of the Ouroboros thesis program, a DOI-pinned lineage of twenty-three prior versions (v1–v23) archived under a Zenodo concept record [17]. Each prior version advanced a piece of the substrate; v23 "Unified Substrate" consolidated them under an *honesty doctrine* (§2) that distinguishes, at every point of assertion, *proven* (kernel-verified, Lean-core axioms only) from *proven under a declared axiom* from *conjectured*.

The v24 advance, stated precisely. v23's strongest Λ-uniqueness result was *conditional* and *gated on a declared project axiom*: `lambda_unique_under_block`, which assumed a named `A6'_block_consistent` idealization. The headline of v24 is that this gate can be removed. The in-tree theorem

`Lutar.Round13.lambda_unique_of_separable`

proves Λ-uniqueness *conditional* on {A1, A2, A3, A5} together with *slice-multiplicativity* (separability), and its trusted base is exactly the Lean core: `#print axioms` = `{propext, Classical.choice, Quot.sound}`— **no project axiom**. It is kernel-clean, CI-green, and merged to `main` at `b910c276`. This *strictly strengthens* v23: the extra hypothesis is now a *checkable structural property of* the candidate aggregator Φ — and one weaker than the old *Factors* premise — discharged through Mathlib and already-proved in-tree lemmas, rather than a disclosed-but-unprovable idealization. We characterize this as an *epistemic upgrade* (§9).

What does *not* change. The single most important honest statement in this program is unchanged and we repeat it without softening: the *unconditional* uniqueness of Λ under {A1–A5} is machine-checked **false**. The in-tree witness `maxAgg_ne_Lambda` exhibits concrete A1–A5 aggregators (`maxAgg` and `min`) that differ from Λ. We therefore label Λ **Conjecture 1** unconditionally — never a theorem — exactly as v23 did. v24 does not, and never will, claim otherwise.

The frontier tier. Alongside the headline theorem, v24 reports a tranche of new *experimental, axiom-free, CI-green* frontier theorems (§6): CF-13 (a deep-equilibrium input-Lipschitz well-posedness margin [4]), CF-17 (a floating-point summation error bound [13]), the Wave-13 results (replay-root completeness, a non-Byzantine quorum shadow, an HM-bottleneck lemma), and the Wave-14 pack — CF-18 (Mādhava/Leibniz alternating-series remainder [21]), CF-19 (a Reed–Solomon MDS distance lower bound [22, 23]), CF-20 (VCG efficiency + truthfulness core [25, 7, 12, 19]), and CF-21 (the Cover–Thomas log-sum + Gibbs inequalities [8]). Every one is `#print axioms` -clean. None is folded into the locked-five.

Reading guide. §2 states the honesty doctrine. §3 recalls the Ouroboros loop. §4 treats Λ and its uniqueness boundary, including the v24 axiom-free theorem (§4.4) with its in-tree lemma chain and worked examples. §5 covers receipts. §6 documents the locked kernel and the frontier families with signatures and axiom-cleanliness. §7 gives the detailed verification status (Waves 11–14, the drift gate, `#print axioms` discipline). §8–§9 make the honest and philosophical case. §10–§11 state the empirical posture and the limitations. Appendices give reproduction and a glossary.

2 The honesty doctrine

The doctrine is load-bearing and is preserved verbatim across all SZL artifacts. It defines tiers of epistemic status; every claim in this paper carries exactly one tag.

[locked / kernel-verified] Proven in Lean 4 with only the trusted core axioms `{propext, Classical.choice, Quot.sound}`; sorry-free; pinned at a fixed commit and part of the *locked* count. Locked-proven = exactly five.

[experimental, axiom-free, CI-green] Proven sorry-free with only the trusted core axioms, CI-green at the recorded commit, but residing in the *experimental* tier and **never folded into the locked count**.

[axiom-gated] Sorry-free *given* an explicitly declared, disclosed idealizing axiom (e.g. hash collision-resistance), with the axiom appearing in the `#print axioms` ledger.

[CI-pending] Signature-checked but not reproducibly green in the wired build; **not** claimed proven.

[machine-checked FALSE] The statement is refuted by a machine-checked counterexample.

[Conjecture 1 – NOT a theorem] A research hypothesis. Λ's unconditional uniqueness lives here.

Non-negotiable rules.

1. **Λ is Conjecture 1 unconditionally** — never claimed proven unconditionally. Unconditional uniqueness under A1–A5 is machine-checked false (`maxAgg_ne_Lambda`). Conditional theorems are claimed only with their hypotheses stated.
2. **Locked-proven = exactly five** {F1, F11, F12, F18, F19} at `c7c0ba17` (749 decl / 14 unique axioms / 163 sorries). This number is never inflated.
3. **The experimental CI-green tier is separate.** At `b910c276`, `main` carries 1323 declarations / 23 axioms (22 unique) / 307 sorries. Nothing in this tier is ever folded into the locked-five.
4. **All axioms are disclosed** via `#print axioms` . No fabricated results; no fake citations.
5. **Supply-chain posture: SLSA L1 honest, L2 roadmap.** We do *not* write "L1+L2 attested", "L2-verified", "L3", "FedRAMP", "Iron Bank", or "CMMC". (§7.4 corrects a v23 overstatement.)
6. **Open sorries are disclosed** (§11); 0 runtime CDN (vendored libraries in-image; only live *data* fetches to public APIs).

3 The Ouroboros loop: bounded self-governance

The first pillar is a self-referential governed computation — the *Ouroboros loop* — in which the system's output feeds its own governance input under a bounded, well-founded recursion: perceive, propose, gate, act, record, and feed the record back. "The loop is the product": the governed recurrence, with its receipts, is the unit of value, not a single inference.

Definition 3.1 (Bounded Ouroboros loop)**.** A loop is *bounded* if it admits a well-founded measure — a ranking function into a well-ordered set that strictly decreases on each governed step, with no infinite descending chain.

Termination is thereby guaranteed; self-reference is rendered coherent by *Tarskian stratification* (a governed object-level computation is audited by a strictly higher meta-level), avoiding the self-verification paradoxes of Gödel and Tarski. The substrate does **not** claim to prove its own consistency; instead proofs are checked by the Lean kernel, the kernel's run by continuous integration, and CI by human sign-off. The bottom turtle — the Lean kernel and its core axioms, the declared cryptographic idealizations, and human sign-off — is finite and disclosed (§A.3).

4 The Lutar invariant Λ and its uniqueness boundary

4.1 Definition and the aggregation axioms

A governed action accumulates evidence along several axes (provenance, policy-compliance, measurability-honesty, moral-grounding, ...), each a sub-score in $[0, 1]$. The governance question is how to combine k sub-scores into one trust verdict.

Definition 4.1 (Lutar invariant)**.** For $x = (x_1, \dots, x_k) \in [0, 1]^k$, the *Lutar invariant* is the equal-weight geometric mean

$$\Lambda_k(x) \;=\; \Big(\prod_{i=1}^{k} x_i\Big)^{1/k}.$$

The geometric mean is the natural aggregator for *ratio-scale* evidence: it commutes with a common rescaling of all axes, and a single zero axis drives the verdict to zero — a disqualifying failure cannot be averaged away. The governance axioms on a candidate aggregator $\Phi : [0, 1]^k \to [0, 1]$ are:

Axiom 1 (Monotonicity, A1)**.** Φ is non-decreasing in each argument.

Axiom 2 (Positive homogeneity, A2)**.** $\Phi(c \cdot x) = c\,\Phi(x)$ for $c > 0$ (scale-covariance).

Axiom 3 (Idempotence / normalization, A3)**.** $\Phi(c, \dots, c) = c$.

Axiom 4 (Boundedness, A4)**.** $\Phi(x) \le \max_i x_i$.

Axiom 5 (Permutation invariance, A5)**.** $\Phi(x_{\sigma(1)}, \dots, x_{\sigma(k)}) = \Phi(x)$ for every permutation σ.

The classical theory (Kolmogorov 1930 [15], Aczél 1948 [1], Hardy–Littlewood–Pólya 1934) confirms that A1–A4 (and even A1–A5) *alone* do not force the geometric mean: the missing ingredient is a *bisymmetry / associativity / replacement* family [26, 2].

4.2 The unconditional refutation: FALSE, machine-checked

It is tempting to claim Λ_k is the *unique* Φ satisfying A1–A5. **This is false, and we prove it false.**

Theorem 4.2 (Refutation of unconditional uniqueness)**.** *[MACHINE-CHECKED FALSE] [CI-GREEN] There exists $\Phi \neq \Lambda_k$ satisfying A1–A5. In particular the max-aggregator* $\mathrm{maxAgg}(x) = \max_i x_i$ *satisfies A1–A5 yet differs from Λ_k: at $(4, 1)$ (on the ratio scale) one has* $\mathrm{maxAgg} = 4$ *while $\Lambda_2(4, 1) = 2$. The minimum aggregator* $\min_i x_i$ *is a second witness.*

Lean reference. The in-tree witness is `Round13.maxAgg_ne_Lambda`. The max function is monotone (A1), positively homogeneous (A2), idempotent (A3), bounded by itself hence $\leq$ max (A4), and symmetric (A5); evaluation at $(4, 1)$ separates it from the geometric mean by `decide`. `#print axioms maxAgg_ne_Lambda` reports Lean-core axioms only. `min` is an analogous `decide`-checked companion. □

This is the epistemic heart of the program: we did not merely *fail* to prove unconditional uniqueness; we *proved its negation.* Consequently Λ remains Conjecture 1 unconditionally (§4.5).

4.3 The old route: uniqueness given factorization, and the A6′ gate

The maximal honestly-true uniqueness statement of the prior lineage was conditional on *factorization*, fully proved in-tree.

Theorem 4.3 (Uniqueness given factorization)**.** *[EXPERIMENTAL, AXIOM-FREE, CI-GREEN] Let Φ satisfy A1–A5 and suppose Φ* factors*: there exist exponents $\alpha_1, \ldots, \alpha_k \geq 0$ with $\Phi(x) = \prod_i x_i^{\alpha_i}$ for all x. Then $\Phi = \Lambda_k$.*

Lean reference and sketch. The Lean term is `lambda_unique_of_factors` (Round-13). Given factorization, idempotence (A3) forces $\sum_i \alpha_i = 1$ and symmetry (A5) forces all α_i equal, whence $\alpha_i = 1/k$ and $\Phi = \Lambda_k$. The exponent collapse uses `NNReal.rpow` arithmetic and a `Finset` induction; the companion `lambda_factors` (axiom-free, CI-green) shows Λ_k itself factors with exponents $1/k$, so the hypothesis is non-vacuous. □

In v23 the substrate's *recommended* route turned the opaque factorization hypothesis into a governance-legible *declared axiom.*

Axiom 6 (Block-consistency / aggregation-invariance, A6′)**.** Aggregating evidence within independent blocks and then across the block results equals aggregating the flattened collection; equivalently, the verdict is invariant to how the auditor partitions evidence into review blocks (after Csátó 2018 [9]).

Theorem 4.4 (v23 conditional uniqueness under declared A6′)**.** *[AXIOM-GATED] [CI-GREEN] Under {A1–A5} together with the single declared axiom `A6'_block_consistent`, Λ_k is the unique normalized aggregator. Its disclosed base is*

`#print axioms lambda_unique_under_block` = `[A6'_block_consistent, propext, Quot.sound, Classical.choice]`

— one declared, non-core project axiom *plus the Lean core.*

Theorem 4.4 was honest but carried a cost: the trusted base included a *project axiom* that could not itself be discharged in-kernel. v24 removes that cost.

4.4 The v24 advance: axiom-free conditional uniqueness (CUT-2)

The v24 headline replaces the declared A6′ idealization with a *checkable structural property* of the candidate aggregator: *slice-multiplicativity* (separability). Informally, Φ is slice-multiplicative if it factors as a product of per-axis slice functions, each of which is multiplicative, normalized, and monotone — properties one can in principle *verify* of a concrete Φ, rather than postulate.

Definition 4.5 (Slice-multiplicativity / separability)**.** $\Phi : [0,1]^k \to [0,1]$ is *slice-multiplicative* if there exist slice functions $f_i : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ such that (sep) $\Phi(x) = \prod_i f_i(x_i)$, (mul) $f_i(s\,t) = f_i(s)\,f_i(t)$, (one) $f_i(1) = 1$, and (mono) each f_i is monotone.

Theorem 4.6 (Axiom-free conditional uniqueness of Λ — the v24 advance)**.** *[EXPERIMENTAL, AXIOM-FREE, CI-GREEN] Let $k > 0$ and let Φ satisfy the Lutar axioms $\{A1, A2, A3, A5\}$. If Φ is slice-multiplicative (Definition 4.5) then $\Phi = \Lambda_k$. The Lean term is `lambda_unique_of_separable` (namespace `Lutar.Round13`); its trusted base is exactly*

`#print axioms lambda_unique_of_separable` = `{propext, Classical.choice, Quot.sound}`,

i.e. ***no project axiom****. It is kernel-clean, CI-green, and merged to `main` at `b910c276`.*

The exact Lean signature is:

```
theorem lambda_unique_of_separable {k : ℕ} (hk : 0 < k)
  (Φ : Aggregator k) (hL : LutarAxioms Φ)
  (f : Fin k → (NNReal → NNReal))
  (hsep : ∀ x, Φ x = ∏ i, f i (x i))
  (hmul : ∀ i s t, f i (s * t) = f i s * f i t)
  (hone : ∀ i, f i 1 = 1)
  (hmono : ∀ i, Monotone (f i)) :
  Φ = Λ k
```

Proof sketch and the in-tree lemma chain. The proof reduces slice-multiplicativity to factorization and then discharges via Theorem 4.3. Concretely:

1. **Per-axis power law.** For each axis i, the slice f_i is multiplicative (mul), monotone (mono), and normalized (one). The in-tree lemma `multiplicative_monotone_isPow_pos` yields an exponent α_i with $f_i(t) = t^{\alpha_i}$ for all $t \neq 0$:

   ```
   multiplicative_monotone_isPow_pos {f} (hf_mul) (hf_mono) (hf_one)
     : ∃α, ∀ t ≠ 0, f t = t^(α : ℝ)
   ```

   Boundary value $f_i(0) = 0^{\alpha_i}$ follows from the idempotent dichotomy (`slice_zero_idem`, `slice_const_one_of_zero_one`).

2. **Exponent equality across axes.** The chosen exponents α_i are forced equal by permutation invariance (A5): applying `Equiv.swap i j` to a two-hot test vector and using the in-tree injectivity lemma `rpow_left_inj_one_lt` ($c^a = c^b \Rightarrow a = b$ for $c > 1$) gives $\alpha_i = \alpha_j$.

3. **Assemble factorization.** Collecting the per-axis power laws yields the `Factors` predicate ($\Phi(x) = \prod_i x_i^{\alpha_i}$).

4. **Discharge.** Apply `lambda_unique_of_factors` (Theorem 4.3); idempotence (A3) and symmetry (A5) collapse the exponents to $1/k$ and conclude $\Phi = \Lambda_k$.

The whole chain `multiplicative_monotone_isPow_pos` → `lambda_unique_of_factors` discharges only through Mathlib and already-verified, sorry-free Round-13 lemmas; no new axiom token is introduced. Hence the trusted base is exactly the Lean core `{propext, Classical.choice, Quot.sound}`. □

Remark 4.7 (Why this is strictly stronger than v23). Slice-multiplicativity is logically *weaker* than the *Factors* premise of Theorem 4.3 (it does not pre-suppose the explicit exponent vector; the exponents are *derived* via the power-law lemma), and it is a *checkable property of* Φ rather than a declared axiom about aggregation. Theorem 4.6 therefore (a) widens the gap between premise and conclusion and (b) removes the project axiom from the trusted base that Theorem 4.4 required. This is the sense in which v24 *strictly strengthens* the v23 conditional uniqueness result.

4.5 Λ is Conjecture 1 — the precise claim structure

We separate three propositions and assert the truth-value of each:

- **(U) Unconditional uniqueness** "Λ is the unique Φ satisfying A1–A5." — **FALSE** (Theorem 4.2).
- **(C) Conditional uniqueness** "Given {A1,A2,A3,A5} + slice-multiplicativity, Λ is unique." — **PROVEN, conditional, axiom-free** (Theorem 4.6).
- **(Conj) The governance conjecture** "The *right* axiomatization of a governed trust aggregator forces Λ." — **Conjecture 1**: a research hypothesis, not a theorem.

Conjecture 1 (The Lutar invariant; never a theorem)**.** *[CONJECTURE 1 – NOT A THEOREM] The equal-weight geometric mean Λ_k is the correct unique trust aggregator for governed AI, in the sense that the governance-correct structural demand (slice-multiplicativity / aggregation-invariance) is the right normative requirement on a governed aggregator. This is an open philosophical claim, not a mathematical theorem; it is* never *asserted as proven, and the unconditional form (U) is* provably false*.*

4.6 Worked examples

- **Disqualifying-zero.** $\Lambda(0.9, 0.9, 0.0) = 0$ — a single failed axis vetoes the verdict, unlike the arithmetic mean (0.6). Trust is not bought back by strong performance elsewhere.
- **Scale-covariance (A2).** $\Lambda(2x) = 2\Lambda(x)$: rescaling the evidence rescales the verdict, the measurement-theoretic objectivity property.
- **maxAgg comparison.** At $(0.8, 0.2)$: $\Lambda = 0.4$ while maxAgg $= 0.8$ — the discriminating instance witnessing Theorem 4.2.
- **Slice-multiplicativity holds for Λ.** Take $f_i(t) = t^{1/k}$: then $\prod_i f_i(x_i) = \left(\prod_i x_i\right)^{1/k} = \Lambda_k(x)$, each f_i is multiplicative, $f_i(1) = 1$, and monotone — so the hypothesis of Theorem 4.6 is realized by Λ itself (non-vacuity).

5 The proof-trail / receipt architecture

Every governed action emits a *receipt*: a canonical record of the action, its inputs, the gate decision, and the Λ sub-scores. The chain entry is $\text{link}_i = H(\text{payload}_i \,\|\, \text{link}_{i-1})$, and a batch commits to a Merkle root.

5.1 Structural correctness (sorry-free)

The replay engine (F1) is deterministic: equal logs replay to equal final state, and the folded state equals the last entry of the explicit replay trace (induction over the log; `#print axioms` reports `propext` only). Hash-chain verification is sound: a verified chain implies all links match. The Merkle inclusion checker is correct-by-construction over an abstract hash. A worked tamper check: altering any $\mathrm{payload}_i$ changes link_i, which propagates to root, so re-verification fails.

5.2 Security under declared idealizations (axiom-gated)

Tamper-evidence, attribution, and inclusion-binding require collision-resistance / unforgeability assumptions. We declare these as named axioms and disclose them, following the ProVerif methodology of abstracting a primitive as an axiom, proving the protocol against it, and discharging the axiom separately for a concrete implementation [5]. The declared axioms include `sha256`, `sha256_collision_resistant`, `node_collision_resistant`, `leaf_collision_resistant`, `compression_collision_resistant`, and `domain_separation`. Each is an injective-oracle abstraction — *not* a proof of cryptographic hardness — disclosed in the `#print axioms` ledger. The architecture retains the full sub-score vector (never letting Λ replace it), gates deny-by-default, and supports byte-identical deterministic replay (F1).

6 The locked kernel and the frontier theorem families

6.1 The locked five (kernel-verified at `c7c0ba17`)

The Lean formalization has a *locked kernel*, pinned at commit `c7c0ba17`: **749 declarations / 14 unique axioms / 163 sorries**. Within this counted scope, exactly **five** governance formulas are proven and locked.

ID	Formula	Status note
F1	Replay determinism	locked
F11	Ayni reciprocity	locked
F12	Kuramoto additive	locked (*additive scaffolding only*, not the full sync theorem)
F18	Reed–Solomon	locked
F19	Bekenstein additive	locked (*additive scaffolding only*, NOT the bound $S \leq 2\pi kRE/\hbar c$)

All other formal work — the experimental pass and every frontier family below — lives in a *separate experimental tier* and is **never** folded into the locked-five.

6.2 The experimental CI-green tier (at `b910c276`)

At `main` commit `b910c276` (Waves 11–14 all merged, CI-green, drift-clean), the experimental tier carries **1323 declarations / 23 axioms (22 unique) / 307 sorries**. The frontier theorems below are all `#print axioms` -clean ($\subseteq$ `{propext, Classical.choice, Quot.sound}`), introduce no new axiom token, and add no `sorry`.

6.3 CF-13 — DEQ input-Lipschitz well-posedness margin

[EXPERIMENTAL, AXIOM-FREE, CI-GREEN] File `Lutar/Innovations/round5/OuroLoopInputLipschitz.lean` (7 declarations). For a contraction with input-Lipschitz constant L_x and contraction modulus $K < 1$, the equilibrium map $z^\star$ satisfies

$$\mathrm{dist}\big(z^\star(x), z^\star(y)\big) \;\le\; \frac{L_x}{1-K}\,\mathrm{dist}(x,y)$$

(`equilibrium_dist_le`, `equilibrium_lipschitz`). Built on Mathlib `ContractingWith`; the result is the well-posedness margin of a deep equilibrium model (Bai–Kolter–Koltun [4]). `#print axioms` = `{propext, Classical.choice, Quot.sound}`.

6.4 CF-17 — floating-point summation error bound

[EXPERIMENTAL, AXIOM-FREE, CI-GREEN] File `Lutar/Khipu/NumericStability.lean` (8 declarations). Under the standard rounding model $\mathrm{fl}(a+b) = (a+b)(1+\delta)$, $|\delta| \le u$ (carried as an explicit hypothesis, not an axiom; Higham [13] §2.2), the recursive sum satisfies $\big|\,\mathrm{recSum}(x,\delta) - \sum_i x_i\,\big| \;\le\; \big((1+u)^{n-1} - 1\big)\sum_i |x_i|$ (`recSum_error_le`). `#print axioms` = `{propext, Classical.choice, Quot.sound}`.

6.5 Wave-13 — replay completeness, quorum shadow, HM bottleneck

[EXPERIMENTAL, AXIOM-FREE, CI-GREEN] (File `Lutar/Wave13/Sweep.lean` and an in-place close.)

- `findReplayRoot_complete` (PRNG, −1 sorry; closed in place via Lean core `List.find?_isSome`): if a replay-root candidate is a member of the candidate list and validates, the search returns `some`. `#print axioms` = `{propext, Quot.sound}`.
- `quorum_agreement_single_valued_vote`: for $n \ge 3f+1$ and two quorums of size $\ge n-f$, single-valued votes agree ($v_1 = v_2$), by nonempty quorum intersection. `#print axioms` = `{propext, Classical.choice, Quot.sound}`. **Honesty:** this assumes each organ votes via a *total function* `voteOf`, so it is an explicitly-labeled *non-Byzantine shadow*, **not** Khipu Conjecture 2 (a faulty organ that equivocates).
- `hm_bottleneck_clean` (HLP harmonic-mean bottleneck): if the harmonic mean falls below a threshold then some coordinate does (contrapositive; pure order/field reasoning). `#print axioms` = `{propext, Classical.choice, Quot.sound}`.

6.6 Wave-14 — CF-18/19/20/21 (nine kernel-clean theorems)

[EXPERIMENTAL, AXIOM-FREE, CI-GREEN] All nine compile clean with `#print axioms` = `{propext, Classical.choice, Quot.sound}`.

- **CF-18 — Mādhava/Leibniz alternating-series remainder** (`Lutar/Wave14/LeibnizRemainder.lean`). For an antitone nonnegative a with alternating partial sums tending to L, one has $\big|\sum_{i<N}(-1)^i a_i - L\big| \le a_N$ (the Lean terms `leibniz_remainder_bound` and `madhava_alt_series_bound_clean`), via Mathlib `Antitone.alternating_series_le_tendsto` even/odd bracketing (Plofker [21]).

- **CF-19 — Reed–Solomon MDS distance lower bound** (`Lutar/Wave14/ReedSolomonDistance.lean`). Two distinct polynomials of degree $< k$ agree on fewer than k points (`agreement_card_lt_of_degree_lt`); lifting via injectivity of the evaluation points gives the bound $n - k + 1 \leq \#\{i : p(\mathrm{pts}_i) \neq q(\mathrm{pts}_i)\}$ (`rs_distance_lower_bound`). This is the *achievability / lower-distance half* of Singleton; the upper bound and full MDS equality remain honest sorries (Reed–Solomon [22], Singleton [23]).
- **CF-20 — VCG efficiency + truthfulness core** (`Lutar/Wave14/VCGEfficiency.lean`). An efficient outcome exists and maximizes social welfare (`exists_efficient_outcome`, `efficientOutcome_maximises`); the truthfulness core is the marginal-contribution monotonicity consequence (`vcg_truthfulness_core`) (Vickrey–Clarke–Groves [25, 7, 12]; Nisan et al. [19]). **Honesty:** the in-tree `MechanismDesign/VCG.lean` references a nonexistent `Finset.argmax` and is unwired; CF-20 is a clean companion, not a patch.
- **CF-21 — Cover–Thomas log-sum + Gibbs** (`Lutar/Wave14/LogSumInequality.lean`). The log-sum inequality (`log_sum_inequality`) and Gibbs' inequality (`gibbs_inequality`) via $\log x \leq x - 1$ (Cover–Thomas [8] Thm 2.7.1 and 2.6.3). **Honesty:** this is the *correctly-stated* DPI core; it does **not** repair the in-tree DPO `klDivergence/pinsker`, which remain false-as-stated for lack of a simplex hypothesis.

7 Verification status, in detail

7.1 The wave ledger (Waves 11–14)

Wave	PR	Adds	Headline
11	#201	24 thm	CF-1/2/3/5 position-aware graph + KV-cache + early-exit + Neyman–Pearson
12	#202	CUT-2 + CF-13 + CF-17	**axiom-free conditional** Λ (`lambda_unique_of_separable`); DEQ; fp-summation; -1 legacy sorry
13	#203	3 thm	replay-root completeness (-1 sorry); non-Byzantine quorum shadow; HM bottleneck
14	#204	9 thm	CF-18 Mādhava/Leibniz; CF-19 RS lower bound; CF-20 VCG; CF-21 log-sum/Gibbs

7.2 The drift gate and the canonical numbers

A numbers-drift gate guards four quantities and the axiom-name set: *declarations*, *axioms_raw*, *axioms_unique*, and *sorries_raw*. At `b910c276` the canonical figures are **1323 / 23 / 22 / 307**. Wave-14 is registered under `EXPERIMENTAL_SCOPES` so its declarations do not move the locked count; the gate is green and drift-clean.

7.3 The `#print axioms` discipline

Every claimed-proven theorem is accompanied by its `#print axioms` ledger. The headline theorem and every frontier family member report `#print axioms` $\subseteq$ `{propext, Classical.`

`choice, Quot.sound}`— the trusted Lean core only — with the sole exception of the explicitly axiom-gated cryptographic results (§5), whose declared idealizations appear in their ledgers. No theorem is stated as proven that CI has not verified at the recorded commit.

7.4 SLSA supply-chain posture — correcting a v23 overstatement

v23 wrote "SLSA L1+L2 attested". **This badge was wrong and is corrected here.** The honest posture is **SLSA L1, with L2 a roadmap item**. We do *not* claim L2-verified, L3, FedRAMP authorization, Iron Bank accreditation, or CMMC certification [20]. The container build pipeline meets L1 provenance expectations; L2 (hosted, authenticated build service with signed provenance) is planned but not yet attested, and is reported as a roadmap item rather than a current state.

8 Why this is groundbreaking — the honest case

Axiom elimination as a first-class result. The standard way to "strengthen" a conditional theorem is to weaken its hypothesis. v24 does something sharper: it *removes a project axiom from the trusted base entirely*. v23's conditional uniqueness depended on `A6'_block_consistent`, a declared idealization that the kernel had to *trust*. v24's `lambda_unique_of_separable` depends on no project axiom at all (`#print axioms` = `{propext, Classical.choice, Quot.sound}`); the only added hypothesis is a *checkable property of* Φ. To our knowledge, exhibiting a governed-AI aggregator uniqueness result whose trusted base is exactly the bare Lean core is new.

A machine-checked boundary, not just a conditional. We did not merely prove a conditional theorem; we proved the *negation* of the unconditional claim (Theorem 4.2). Establishing exactly which extra assumption is required — and that without it the claim fails — converts a routine conditional into a *boundary* result: Theorem 4.6 is demonstrably close to the maximal true statement of its kind, and the boundary itself is machine-checked.

Disclosed-axiom discipline at scale. Formal verification of security-critical systems is well established (seL4 is the canonical machine-checked OS kernel [14]); applying a disclosed-axiom, kernel-checked discipline to an AI governance aggregator and its receipt architecture — with a `#print axioms` ledger for *every* claim and a hard separation between a locked five and an experimental tier — is, as far as we can determine, new.

8.1 Honest comparison to prior art

Prior art	What it provides	How v24 differs (honestly)
Verifiable-claims governance [6, 18]	The argument that AI claims should be externally verifiable.	A concrete substrate with a kernel-checked core; does not claim to solve verifiability in general.
seL4 / formal OS [14]	Machine-checked OS-kernel correctness; the gold standard.	Verifies a governance aggregator + receipt layer; far smaller in scope and maturity; no seL4-level coverage claimed.
Certificate Transparency [16]	Merkle-based tamper-evident logs in production.	Reuses the Merkle discipline; formalizes the binding property in Lean under declared axioms.
Functional-equation aggregation [1, 3, 9]	Uniqueness characterizations of the geometric mean.	Formalizes an *axiom-free* conditional uniqueness (slice-multiplicativity) and machine-checks the unconditional refutation.
SLSA framework [20]	A supply-chain integrity ladder (L1–L4).	Honestly attests L1; L2 is a roadmap item; L3 / FedRAMP / Iron Bank / CMMC explicitly *not* claimed.

Honest summary. The individual mathematical facts (geometric-mean characterizations, Kraft, Shannon, Byzantine quorums, Reed–Solomon, VCG, log-sum) are classical. The contribution of v24 is their unification into a governed-AI trust substrate, their kernel-checked instantiation, and above all the *axiom elimination* on the central uniqueness result combined with the *honesty architecture* — locked counts, disclosed axioms, and a machine-checked refutation guarding the one tempting overclaim.

9 Philosophical foundations: slice-multiplicativity as an epistemic upgrade

9.1 From a declared axiom to a checkable property

A trusted base that contains a project axiom asks the auditor to *believe* something the kernel cannot check. A6′ (block-consistency) was such an axiom: governance-legible, published, and defensible, but ultimately *postulated*. Slice-multiplicativity is different in kind. It is a structural *property of the candidate aggregator* Φ — of the form "Φ factors into multiplicative, normalized, monotone slices" — that one can in principle *verify* of a concrete Φ by inspecting its definition. Replacing a postulate the auditor must trust with a property the auditor can check is, on any reliabilist account of justification [11], an *epistemic upgrade*: the warrant for the conclusion no longer routes through an unverifiable idealization.

9.2 When is a conditional legitimate?

Four conditions, all met by Theorem 4.6: (i) *independently-motivated antecedent* — separability is the standard hypothesis under which multiplicative aggregators are characterized [2]; (ii) *non-vacuous instantiation* — Λ itself is slice-multiplicative (§4.6), so the conditional is not vacuously true; (iii) *boundary established* — without the hypothesis, uniqueness fails (Theorem 4.2); (iv) *disclosed at point of use* — the `#print axioms` ledger shows the bare core. The residual gap — *is slice-multiplicativity the philosophically correct demand on a governed aggregator?* — is genuinely open, which is exactly why Λ remains Conjecture 1.

9.3 Slice-multiplicativity vs. block-consistency

Both A6′ and slice-multiplicativity are sufficient to pin Λ. The relevant difference is epistemic, not merely logical. A6′ entered the trusted base as a named axiom; slice-multiplicativity enters as a *hypothesis discharged through the kernel*. The former widens the trusted computing base; the latter does not. We therefore regard the v24 route as dominating the v23 route along the one axis that matters to an auditor: *how much must be trusted that cannot be checked*.

9.4 Steelman objections (with honest rebuttals)

1. *"Slice-multiplicativity is the geometric mean in costume."* — It is strictly weaker than the explicit *Factors* premise and does not name the exponents; the power-law lemma *derives* them. The residue is a defensible normative choice, labeled as Conjecture 1.
2. *"Motte-and-bailey."* — The conditional theorem and the bald "Λ uniqueness" are labeled and separated in the artifacts themselves; (U) is stated false.
3. *"Idealized crypto axioms make guarantees illusory."* — Conceded and disclosed; receipt guarantees are conditional on the same assumptions TLS and Certificate Transparency already rely on [16].
4. *"Arrow: aggregation is impossible."* — Arrow concerns ordinal preference aggregation; Λ aggregates cardinal ratio-scale evidence about one object, outside Arrow's IIA impossibility.
5. *"Gödel/Tarski: self-auditing is incoherent."* — Conceded as a limit; the substrate uses bounded recursion verified by an external stratified stack, not self-verification.

10 Empirical posture (measured, not proven)

The figures below are *measured*, reported as empirical percentiles over archived runs, with no theorem claimed. Per-receipt build latency $\approx 11.5\,\mu s$ (p50) / $50.7\,\mu s$ (p99); verification $\approx 10.4\,\mu s$ (p50). Nine-axis $\Lambda \approx 3.12\,\mu s$ (base) / $3.29\,\mu s$ (composed). On a prior platform run of 24,800 HTTP calls, governance overhead was 0.49–0.59 ms per route (p50), 1.27 ms (p99). A deterministic 5× replay yields a byte-identical root.

Operation	**p50**	**p99**	**What is measured**
Receipt build	$11.5\,\mu s$	$50.7\,\mu s$	Canonicalize + hash + chain + persist
Receipt verify	$10.4\,\mu s$	—	Chain re-check + root comparison
Λ_9 (base)	$3.12\,\mu s$	—	Nine-axis geometric mean, isolated
Λ_9 (comp.)	$3.29\,\mu s$	—	Within a composed gate evaluation
Gov. overhead / route	0.49–0.59 ms	1.27 ms	Governed minus ungoverned, 24,800 calls

These are wall-clock timings collected in-process with a monotonic nanosecond clock over archived runs; no confidence intervals or cross-run variance are claimed.

11 Limitations and honesty

- **Λ is Conjecture 1 unconditionally.** Unconditional uniqueness under A1–A5 is false (Theorem 4.2). Only the conditional theorem holds, and whether slice-multiplicativity is the philosophically correct demand is open. Λ is never claimed proven unconditionally.
- **The locked count is exactly 5.** {F1,F11,F12,F18,F19} at `c7c0ba17` (749/14/163). The experimental CI-green tier (1323/23/22/307 at `b910c276`) is separate and never folded in.
- **Byzantine BFT safety stays Khipu Conjecture 2 (open).** A faulty organ can equivocate; `ubuntu_quorum_safety` is untouched. The Wave-13 `quorum_agreement_single_valued_vote` is an explicitly non-Byzantine shadow, not a resolution.
- **DPO `klDivergence_nonneg / pinsker` are false as stated** (no simplex constraint on `PolicyParam`); they remain honest declared axioms. CF-21 proves the correctly-stated log-sum/Gibbs core but does *not* repair them.
- **Open sorries remain.** The locked kernel carries 163 sorries; the experimental scope carries 307 raw sorries. RS Singleton upper bound / full MDS equality, the in-tree VCG file, Bekenstein/Bousso IQ-gate stubs, and the hard sites (Putnam analysis, xoshiro period via $GF(2)^{256}$ primitivity, Hoeffding–Azuma, Brouwer/cohomology, gauge symmetry, path-integral convergence) stay honest sorries with references intact.
- **SLSA L1 honest, L2 roadmap.** Not L2-verified, L3, FedRAMP, Iron Bank, or CMMC (§7.4).
- **Declared idealizations.** Tamper-evidence, attribution, and Merkle binding hold only under declared collision-resistance / unforgeability axioms, disclosed in every `#print axioms` ledger.
- **Scope.** The substrate verifies a governance aggregator and receipt layer, not an end-to-end AI system; it is far smaller in scope and maturity than a fully verified OS kernel.

12 Conclusion

v24 advances the Ouroboros program by a single, sharply-stated step: the central trust aggregator's conditional uniqueness, which in v23 rested on a declared block-consistency axiom, now rests on *no project axiom at all*. The in-tree `lambda_unique_of_separable` proves Λ-uniqueness under {A1,A2,A3,A5} plus slice-multiplicativity with `#print axioms = {propext, Classical.choice, Quot.sound}`, kernel-clean and CI-green at `b910c276`, and the extra hypothesis is a checkable property of Φ rather than an idealization. The unconditional claim remains machine-checked *false*, so Λ stays Conjecture 1, exactly as before.

> **The one-line thesis.** *v24 strictly strengthens the v23 conditional uniqueness of the Lutar invariant by eliminating the declared block-consistency axiom: Λ-uniqueness is now proven conditional on {A1,A2,A3,A5} plus a checkable slice-multiplicativity property with a trusted base of exactly the bare Lean core, while its unconditional uniqueness remains machine-checked false — so the axiom-free conditional theorem is the maximal true statement and the invariant is honestly labeled Conjecture 1, never a theorem.*

The contribution is not a new theorem alone but a new *standard*: axiom elimination on the load-bearing result, locked counts held separate from an experimental tier, disclosed axioms, and a machine-checked refutation guarding the one tempting overclaim. We submit that this calibrated honesty is precisely what a regulator, an auditor, or a defense customer should require.

A Reproducibility and artifact manifest

A.1 Pinned commits and artifacts

Component	Pin	Contents / status
`main` (Waves 11–14 merged)	`b910c276`	1323 decl / 23 axioms (22 unique) / 307 sorries; CI-green
Locked kernel	`c7c0ba17`	749 / 14 / 163; locks {F1,F11,F12,F18,F19}
CUT-2 axiom-free Λ	in `b910c276`	`Round13/LambdaSeparable.lean`
Concept DOI (always-latest)	—	10.5281/zenodo.19944926
Repository	—	github.com/szl-holdings/lutar-lean

A.2 How to verify

```
# 1. Kernel-check the merged library at the pinned commit
$ git checkout b910c276 && lake build                # expect: green, no errors

# 2. Disclose the trusted axiom base of the v24 headline theorem
$ #print axioms Lutar.Round13.lambda_unique_of_separable
=> [propext, Classical.choice, Quot.sound]                      # NO project axiom

# 3. Confirm the unconditional claim is refuted
$ #print axioms maxAgg_ne_Lambda                            # Lean-core axioms only

# 4. Verify the locked kernel separately
$ git checkout c7c0ba17 && lake build               # locks F1,F11,F12,F18,F19
```

A.3 Trusted computing base

The bottom turtles, stated once: (i) soundness of the Lean 4 kernel and its core axioms `{propext, Classical.choice, Quot.sound}`[10, 24]; (ii) the declared cryptographic idealizations (`sha256_collision_resistant` and the Merkle collision-resistance / domain-separation axioms), used only in the receipt-security theorems; and (iii) human sign-off gating CI. The v24 headline theorem `lambda_unique_of_separable` depends on *none* of the cryptographic idealizations and on *no* project axiom — only on (i).

B Notation and glossary

Symbol / term	Meaning
$\Lambda_k(x)$	The Lutar invariant: equal-weight geometric mean $(\prod_i x_i)^{1/k}$ on $[0,1]^k$
Φ	A generic candidate aggregator $[0,1]^k \to [0,1]$ satisfying some subset of A1–A5
`maxAgg`, `min`	The machine-checked A1–A5 witnesses that A1–A5 do not pin Λ (Thm 4.2)
A1–A5	Monotonicity, positive homogeneity, idempotence, boundedness, permutation-invariance
A6′	Block-consistency: the v23 *declared* axiom (replaced in v24)
slice-multiplicativity	The v24 *checkable* hypothesis: $\Phi = \prod_i f_i(x_i)$ with each f_i multiplicative, normalized, monotone
`lambda_unique_of_separable`	The v24 headline: axiom-free conditional Λ-uniqueness
link_i, root	The i-th hash-chain entry and the Merkle commitment to a receipt batch
sorry-free	A Lean development with no `sorry`; the kernel checks every step
CI-green	The real CI `lake build` (a full kernel check) passes at the pinned commit
axiom-gated	Sorry-free given an explicitly declared, disclosed idealizing axiom
`#print axioms`	Lean command listing the trusted axiom base of a theorem
locked-five	The exactly five kernel-verified formulas {F1,F11,F12,F18,F19} at `c7c0ba17`
experimental tier	The separate CI-green corpus (1323/23/307 at `b910c276`), never folded into the locked-five
Conjecture 1	The open claim that Λ is the correct unique governed aggregator; never a theorem

References

[1] János Aczél. On mean values. *Bulletin of the American Mathematical Society*, 54(4):392–400, 1948. https://eudml.org/doc/296298.

[2] János Aczél. *Lectures on Functional Equations and Their Applications*. Academic Press, 1966. Bisymmetry / associativity characterizations of means.

[3] János Aczél and Thomas L. Saaty. Procedures for synthesizing ratio judgements. *Journal of Mathematical Psychology*, 27(1):93–102, 1983.

[4] Shaojie Bai, J. Zico Kolter, and Vladlen Koltun. Deep equilibrium models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019. arXiv:1909.01377, https://arxiv.org/abs/1909.01377.

[5] Bruno Blanchet et al. Automatic verification of transparency protocols. arXiv:2303.04500, https://arxiv.org/abs/2303.04500, 2023.

[6] Miles Brundage, Shahar Avin, Jasmine Wang, et al. Toward trustworthy AI development: Mechanisms for supporting verifiable claims. Technical Report arXiv:2004.07213, arXiv, 2020. https://arxiv.org/abs/2004.07213.

[7] Edward H. Clarke. Multipart pricing of public goods. *Public Choice*, 11:17–33, 1971.

[8] Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. Wiley-Interscience, 2 edition, 2006. §2.6–2.8, log-sum inequality (Thm 2.7.1), Gibbs' inequality (Thm 2.6.3); ISBN 978-0-471-24195-9.

[9] László Csátó. Characterization of the row geometric mean ranking with a group consensus axiom. *Group Decision and Negotiation*, 27(6):1011–1027, 2018. arXiv:1706.07256, https://arxiv.org/abs/1706.07256.

[10] Leonardo de Moura and Sebastian Ullrich. The Lean 4 theorem prover and programming language. CADE-28; https://leanprover.github.io, 2021.

[11] Alvin I. Goldman. What is justified belief? (process reliabilism). Reliabilist Epistemology, Stanford Encyclopedia of Philosophy, https://plato.stanford.edu/entries/reliabilism/, 1979.

[12] Theodore Groves. Incentives in teams. *Econometrica*, 41(4):617–631, 1973.

[13] Nicholas J. Higham. *Accuracy and Stability of Numerical Algorithms*. Society for Industrial and Applied Mathematics (SIAM), 2 edition, 2002. Floating-point summation error model, §2.2–§4.2.

[14] Gerwin Klein, Kevin Elphinstone, Gernot Heiser, et al. seL4: Formal verification of an OS kernel. In *Proc. ACM SOSP*, 2009.

[15] Andrey N. Kolmogorov. Sur la notion de la moyenne. Atti della Accademia Nazionale dei Lincei 12:388–391, 1930. Quasi-arithmetic (Kolmogorov–Nagumo) means; see https://en.wikipedia.org/wiki/Quasi-arithmetic_mean.

[16] Ben Laurie, Adam Langley, and Emilia Kasper. RFC 6962: Certificate transparency. https://www.rfc-editor.org/info/rfc6962/, 2013.

[17] Stephen P. Lutar. The ouroboros thesis / unified substrate — concept record (always-latest). Zenodo concept DOI, 2026. https://doi.org/10.5281/zenodo.19944926.

[18] National Telecommunications and Information Administration. AI accountability policy report: Proof of claims and trustworthiness. https://www.ntia.gov/issues/artificial-intelligence/ai-accountability-policy-report, 2024.

[19] Noam Nisan, Tim Roughgarden, Éva Tardos, and Vijay V. Vazirani. *Algorithmic Game Theory*. Cambridge University Press, 2007. Ch. 9, VCG mechanisms; ISBN 978-0-521-87282-9.

[20] Open Source Security Foundation. SLSA: Supply-chain levels for software artifacts (levels 1–4). https://slsa.dev, 2023.

[21] Kim Plofker. *Mathematics in India*. Princeton University Press, 2009. §7.4, Mādhava–Leibniz series and remainder; ISBN 978-0-691-12067-6.

[22] Irving S. Reed and Gustave Solomon. Polynomial codes over certain finite fields. *Journal of the Society for Industrial and Applied Mathematics*, 8(2):300–304, 1960.

[23] Richard C. Singleton. Maximum distance q-nary codes. *IEEE Transactions on Information Theory*, 10(2):116–118, 1964.

[24] The Mathlib Community. The Lean mathematical library. Proc. CPP 2020; `https://github.com/leanprover-community/mathlib4`, 2020.

[25] William Vickrey. Counterspeculation, auctions, and competitive sealed tenders. *The Journal of Finance*, 16(1):8–37, 1961.

[26] Wikipedia. Quasi-arithmetic mean (kolmogorov–nagumo–de finetti characterization). `https://en.wikipedia.org/wiki/Quasi-arithmetic_mean`, 2024.

Signed-off-by: Stephen P. Lutar Jr. Co-authored with the SZL full-stack PhD team (mathematics, scientific writing, physics, CS, ML, philosophy of mathematics).

Honesty doctrine preserved verbatim: Λ is Conjecture 1 unconditionally (never a theorem); the v24 conditional theorem `lambda_unique_of_separable` holds under {A1,A2,A3,A5} + slice-multiplicativity with `#print axioms` = {`propext, Classical.choice, Quot.sound`}(no project axiom); locked/proven = exactly five; the experimental CI-green tier is separate and never folded in; declared axioms disclosed in every `#print axioms` ledger; SLSA L1 honest, L2 roadmap (not L2-verified, L3, FedRAMP, Iron Bank, or CMMC). No fabricated results, no fake citations, 0 runtime CDN.